

Daniel Zavorotny

GPU & Machine Learning Engineer
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EXPERIENCE

Graduate Research Assistant

UC Davis VIDI Lab

Mar 2023 – Present

- Built a multi-resolution GPU cache into a neural (INR) volume renderer, eliminating redundant per-frame network inference — an average 5x ray-marching speedup over the prior state of the art on a single RTX 3080Ti.
- Compressed a 0.96 TB fluid-simulation dataset into a 150 MB neural model (6444:1, 35 dB PSNR) and rendered it interactively at over 90 FPS on desktop hardware.
- Designed the cache's priority scheduling and preloading, cutting cold-start misses and stabilizing framerate across ten test volumes from 34 GB to 275 GB.
- Extended CUDA ray-marching and path-tracing pipelines with cache-aware sampling and stochastic level-of-detail transitions, removing visible artifacts at detail boundaries at negligible cost.
- Rebuilt a neural radiance cache around ray-traced 3D Gaussians (CUDA, OptiX) and benchmarked it head-to-head against neural and Gaussian-splat cache baselines from a single binary — roughly 10x smaller cache footprint at matched image quality; manuscript in revision.
- Profiled CUDA renderers with NVTX instrumentation and per-frame timing and memory telemetry, attributing frame cost across pipeline stages to pin down cache bottlenecks.

Computer Science Teaching Assistant

University of California, Davis

- **Computer Architecture:** co-designed exams, led grading, held office hours on CPU and circuit design. Jan – Apr 2026
- **Computer Vision:** led discussion sections on ML methods for vision; graded coursework. Sep – Dec 2025

Technician & Financial Officer

VZ Plumbing Inc.

Jan 2021 – Present

- Own estimating, invoicing, permitting, and contract documentation for a licensed plumbing contractor — the operational experience that directly informed the design of Meliora.

PUBLICATIONS

From Cluster to Desktop: A Cache-Accelerated INR Framework for Interactive Visualization of Tera-Scale Data

EGPGV 2025 (Eurographics Symposium on Parallel Graphics and Visualization) — first author, Honorable Mention.

PROJECTS

Meliora

Construction management platform (melioraos.com)

2026 – Present

- Turned firsthand construction-industry experience into a working product: designed and built the platform solo — 50k+ lines of code — and took it from first commit to a general contractor using it in private beta in 47 days.
- Developed a disciplined, agent-driven engineering process along the way — design documents as the source of truth, automated quality gates, tests pinned to a hand-written financial specification, adversarial code review — sustaining a release every few days for a private-beta customer.

Okibi

Task management application (okibi.app)

2026 – Present

- Designed and shipped a personal task manager end to end — authentication, cloud database, and two-way Google Calendar sync built defensively so a sync conflict can never destroy user data — live at okibi.app after 11 releases in its first seven weeks.

SKILLS

GPU & Rendering: CUDA, OptiX, 3D Gaussian splatting, differentiable rendering, GPU profiling (Nsight / NVTX)

Programming: C, C++, Python, CUDA

Machine Learning: PyTorch, computer vision, implicit neural representations, hash-grid encodings

Systems & Tooling: CMake, Docker, Linux, Git, Conda, Claude Code, GitHub Copilot

Application: Next.js, React, Node.js, Supabase, Postgres

EDUCATION

MS, Computer Science

University of California, Davis — GPA 3.95

2025 – 2026

BS, Computer Science

University of California, Davis — GPA 3.78

2021 – 2023

AS, Physics, Mathematics & Economics

American River College — GPA 3.95

2019 – 2021